

Step-by-Step Building Tiny Bootable C shell

Requirement:

Programming language: C for shell coding, RISC-V assembly code for boot loader

Hardware: We will emulate a hardware RISC-V CPU using **qemu**

Target Operating system: This tutorial is design for ubuntu terminal

Step1 : Install qemu risc-v on your ubuntu terminal. Type the following in you terminal:

```
sudo apt install gcc-riscv64-unknown-elf qemu-system-misc
```

It will install gcc for riscv 64 bit and qemu in your system

Step 2: Project layout: create a directory named riscv-shell under it copy following files:

1. boot.S
2. shell.c
3. uart.c
4. uart.h
5. linker.ld
6. Makefile

Booting your Shell.c in qemu-riscV

```
$cd riscv-shell
```

```
$make
```

```
$make run //It will run the qumu which will boot your Shell.c
```